SellsPark

Node.js Developer Assessment – Fresher Level

Total Time: 50 Minutes, **Total Marks:** 100

Full Name _____

Email ID _____

Contact Number _____

Instructions

1. Read all questions carefully before attempting.
2. Partial marks may be awarded for correct logic even if syntax is imperfect.
3. Do not use the internet or any external code reference.
4. Write neatly and clearly — illegible answers may not be evaluated.

Section A: Node.js & JavaScript (10 Questions – 35 Marks)

1. What is the difference between `setTimeout()` and `setImmediate()` in Node.js?
2. Explain the **event loop** in Node.js and its main phases.
3. What are **callbacks** and how are they different from **Promises** and **async/await**?
4. Define **middleware** in Express.js. Write a small example.
5. Write a Node.js script to create an HTTP server that returns “Hello World” on port 3000.
6. Difference between `require()` and `import`.
7. How do you handle **uncaught exceptions** and **unhandled promise rejections**?
8. What are **streams** in Node.js? Give a use case.
9. Write an Express.js route that connects to a database (MongoDB or MySQL) and returns all records from a table/collection?
10. What is **CORS** and how do you enable it in Express.js?

Section B: C++ Basics (5 Questions – 15 Marks)

11. Difference between **stack** and **heap** memory.
12. Write a program to find the largest of three numbers.
13. Difference between **struct** and **class** in C++.
14. Explain **constructor** and **destructor** with examples.
15. What will be the output of the following code and why?

```
int a = 5;
cout << ++a * a++;
```

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Section C: Networking Fundamentals (5 Questions – 15 Marks)

16. Define **IP address** and how it differs from a **MAC address.**
17. What is the function of **DNS**?
18. Define **ports.** Mention default ports for HTTP, HTTPS, and SSH.
19. Differentiate between **TCP** and **UDP** with examples.
20. What does the **ping** command do?

Section D: Linux Commands (3 Questions – 10 Marks)

21. State the use of following commands:
    a) `grep`
    b) `chmod`
    c) `cat`
22. Command to list all files (including hidden ones) in a directory:
23. Difference between `>` and `>>` in Linux redirection.

Section E: Aptitude & Logic (3 Questions – 10 Marks)

24. If 15 pens cost the same as selling price of 12 pens, find profit percentage.
25. A 120m train passes a pole in 6s. Find its speed in km/hr.
26. Find the missing number: 2, 6, 12, 20, 30, ?

Section F: Data Structures & Algorithms (2 Questions – 8 Marks)

27. Write a function to reverse a string without using built-in reverse methods.
28. Explain **time complexity** and **space complexity** with examples.

Section G: Tech Awareness & Trends (2 Questions – 7 Marks)

29. How do you stay updated with latest software and Node.js trends?
30. What is **containerization (Docker)** and why is it useful in development?